

Study Questions: Communication

1. How does agent A model agent B in the coupled inference framework? Write the generative model.
2. Write agent A's free energy F_A when it includes hidden states of agent B.
3. What is coupled inference? How does it differ from single-agent inference?
4. Define the joint free energy F_{joint} for a system of N agents.
5. How does the interaction term in joint free energy measure the alignment of generative models?
6. What is generalized synchrony? Define it formally.
7. What is the synchronization manifold? Under what conditions does it become an attractor?
8. How does coupling strength relate to the stability of generalized synchrony?
9. What is the relationship between mutual information and communication success?
10. How does the concept of common ground emerge from the alignment of generative models?
11. Derive the belief update equation for agent A when it observes the actions of agent B.
12. How does turn-taking in conversation relate to alternating free energy minimization?
13. What is the mathematical relationship between theory of mind and hierarchical modeling of another agent?
14. How does the concept of Nash equilibrium relate to joint free energy minimization?
15. Can communication fail due to incompatible generative models? Formalize this mathematically.
16. How does the exchange of cultural practices relate to shared prior distributions?
17. What is the role of shared environmental context in reducing the complexity of inter-agent inference?
18. How does Active Inference model deception? (Hint: an agent acts to create prediction errors in the other agent's model.)
19. Derive the conditions under which consensus (all agents converging to the same belief) emerges from coupled inference.
20. How does the mathematics of coupled oscillators (Kuramoto model) relate to the synchronization of Active Inference agents?